

EXERCISE/NUMERICAL ANALYSIS

Q1. The total charge entering a terminal is given by $q = 5t \sin(4\pi t)$ mC. Calculate the current at $t = 0.5$ s.

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Answer:

$$\begin{aligned} i &= \frac{dq}{dt} = \frac{d}{dt} (5t \sin 4\pi t) \text{ mC} / \text{s} \\ &= (5 \sin 4\pi t + 20\pi t \cos 4\pi t) \text{ mA} \end{aligned}$$

At $t = 0.5$,

$$i = 5 \sin 2\pi + 10\pi \cos 2\pi = 31.42 \text{ mA}$$

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Q2. Determine the total charge entering a terminal between $t = 1\text{s}$ and $t = 2\text{s}$ if the current passing the terminal is $i = (3t^2 - t) \text{ A}$.

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Answer:

$$Q = \int_{t=1}^2 i \, dt = \int_1^2 (3t^2 - t) \, dt = \left(t^3 - \frac{t^2}{2} \right) \Big|_1^2$$

$$= (8 - 2) - \left(1 - \frac{1}{2} \right) = 5.5 \text{ C}$$

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Q3. An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

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Answer:

The total charge is

$$\Delta q = i \times \Delta t = 2 \times 10 = 20 \text{ C}$$

The voltage drop is

$$v = \frac{\Delta w}{\Delta q} = \frac{2.3 \times 10^3}{20} = 115 \text{ V}$$

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Q4. To move charge q from point b to point a requires 25 J. Find the voltage drop v_{ab} (the voltage at a positive with respect to b) if: (a) $q = 5$ C, (b) $q = -10$ C.

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Answer:

(a)
$$v = \frac{\Delta w}{\Delta q} = \frac{25}{5} = 5V$$

(b)
$$v = \frac{\Delta w}{\Delta q} = \frac{25}{-10} = -2.5V$$

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Q5. How much energy (in both kWh and Joules) does a 100-W electric bulb consume in two hours?

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Answer:

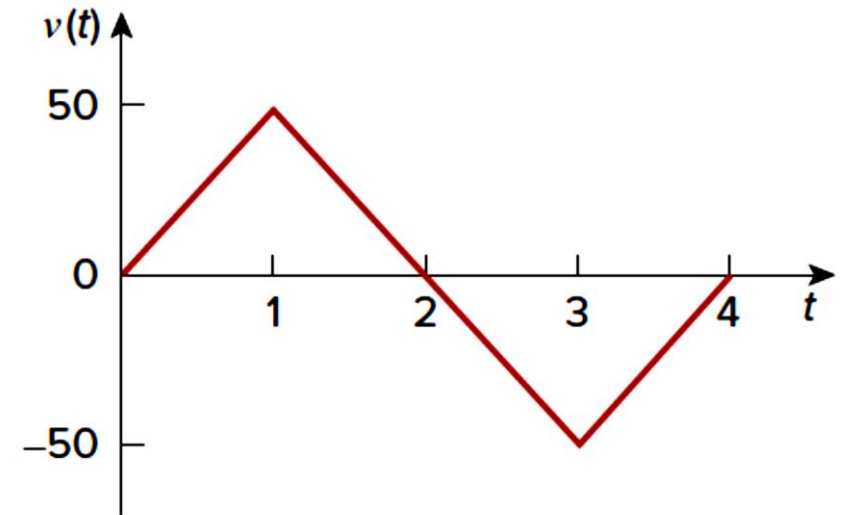
$$\begin{aligned} w &= p \times t = 100(\text{ watt }) \times 2(\text{ h }) \times 60(\text{ min/ h }) \times 60(\text{ sec/ min }) \\ &= 720,000 \text{ Joule} = 720 \text{ kJ} \end{aligned}$$

This is the same as

$$w = pt = 100 \times 2 = 200 \text{ watt} - \text{hour}$$

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Q6. Determine the current as a function of time through a $200\text{-}\mu\text{F}$ capacitor whose voltage is shown in the figure



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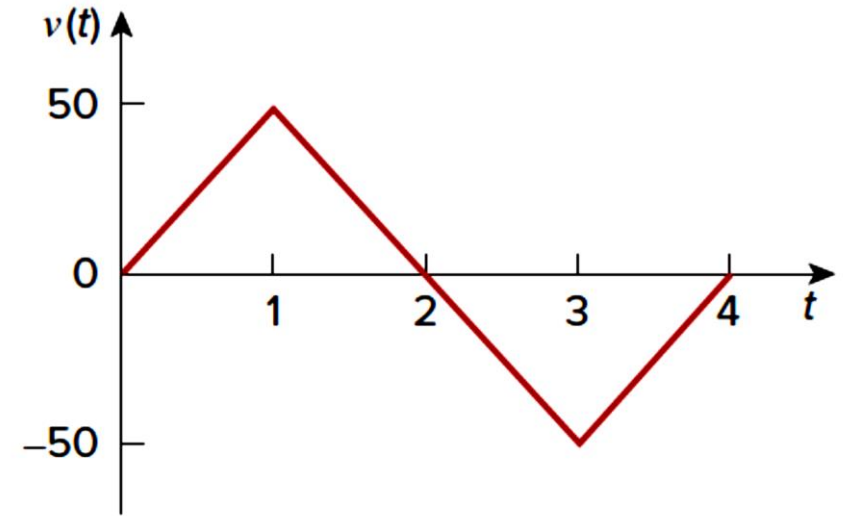
Q6. Determine the current as a function of time through a 200- μ F capacitor whose voltage is shown in the figure

Answer:

The voltage waveform can be described

mathematically as,

$$v = \begin{cases} 50t & 0 < t < 1 \\ 100 - 50t & 1 < t < 3 \\ -200 + 50t & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$



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Since $i = C dv/dt$ and $C = 200 \mu F$, we take derivative of v to obtain,

$$i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 10 \text{ mA} & 0 < t < 1 \\ -10 \text{ mA} & 1 < t < 3 \\ 10 \text{ mA} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Thus, current waveform is as shown in figure.

